PROJECT REPORT

Name: **S.Rithanya**

Registration no : **25BOE10067**

Title: **Aqua pH monitoring system**

Introduction

This project implements a Python-based simulation of an aquarium pH monitoring system. The system generates random pH values within typical aquarium ranges, evaluates their safety for aquatic life, and provides maintenance recommendations. It demonstrates fundamental programming concepts applied to real-world environmental monitoring.

Objectives

- Simulate realistic aquarium pH monitoring
- Implement conditional logic for safety evaluation
- Generate actionable recommendations based on pH levels
- Create a user-friendly console interface
- Model environmental monitoring principles

System Requirements

Safe pH Range

- Optimal Range: 6.8 - 8.0 (Safe for most freshwater aquatic species)
- Unsafe (Acidic): < 6.8

- Unsafe(Alkaline): > 8.0

Algorithm:

1. Generate random pH: uniform(6.0, 8.5), rounded to 1 decimal

2. Evaluate status:

   IF 6.8 ≤ pH ≤ 8.0 → "SAFE"

   ELSE → "UNSAFE"

3. Display formatted output

4. Provide recommendations based on status and pH value

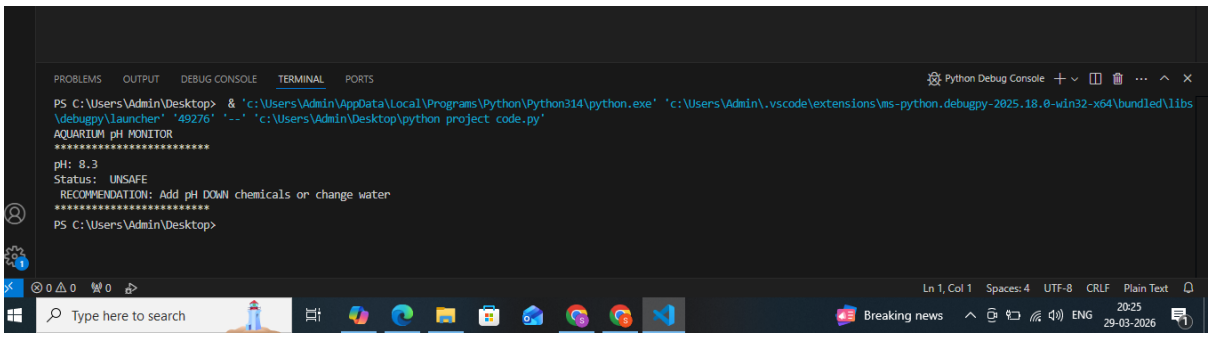
Code:

```
import random

#ph generation

ph = round(random.uniform(6.0, 8.5), 1)

#safety evaluation

if 6.8 <= ph <= 8.0:

    status = "SAFE"

else:

    status = " UNSAFE"

#output

print("AQUARIUM pH MONITOR")

print("*" * 25)

print(f"pH: {ph}")

print(f"Status: {status}")
```

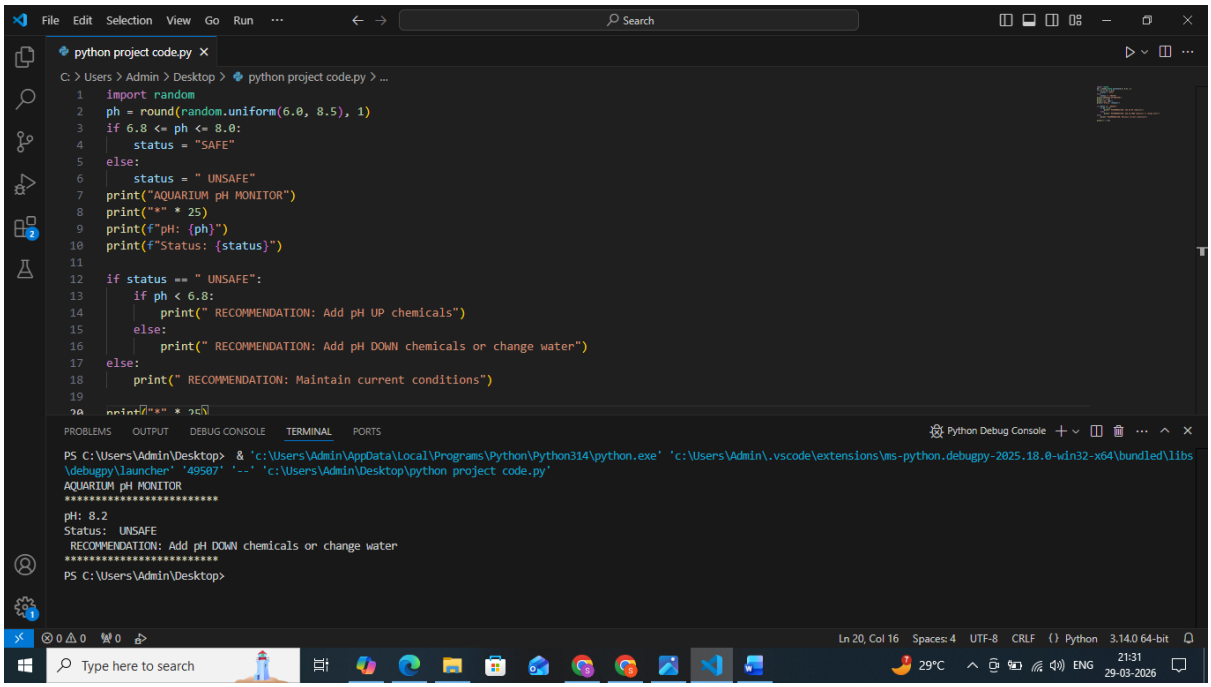
```
if status == " UNSAFE":
    if ph < 6.8:
        print(" RECOMMENDATION: Add pH UP chemicals")
    else:
        print(" RECOMMENDATION: Add pH DOWN chemicals or change water")
else:
    print(" RECOMMENDATION: Maintain current conditions")

print("*" * 25)
```

Sample outputs

1.



2.



Programming concepts demonstrated

- Random number generation
- Conditional statements
- Boolean logic
- Numeric rounding
- String formatting

Limitations:

- Single parameter monitoring (pH only)
- No persistent data storage
- Console-based interface only
- Static safe range (not species-specific)

Future Enhancements:

- Multi-parameter monitoring (temperature, ammonia, nitrates)

- Data logging to CSV files
- Graphical user interface (Tkinter/PyQt)
- Species-specific pH ranges
- Trend analysis and alerts
- Web-based dashboard

Conclusion:

An Aquarium pH monitoring system successfully demonstrates how basic python programming can model real world environmental monitoring applications. The implementation robust ,user friendly and provides accurate recommendations based on real life science applications.